

$\text{\LaTeX}2_{\epsilon}$ EDPSCI Document Class

Author Instructions

EDP Sciences

1. Introduction

The documentation in the EDP Sciences tool package is not intended to be a general introduction to $\text{\LaTeX}2_{\epsilon}$ or $\text{\TeX}$. Instead, the EDP Sciences EDPSCI tool package has been set up for those who are familiar with the basics of $\text{\LaTeX}$. The EDPSCI document class and its special features were designed to facilitate the preparation of Research article to EDP's style requirements.

If in this tool package we refer to standard tools or packages that are not installed on your system.

The components of the EDPSCI tool package are:

- The *EDP $\text{\LaTeX}$ class* `edpsci.cls` with preset class options, packages and coding examples.
Tip: Copy these files to your working directory, run $\text{\LaTeX}2_{\epsilon}$ and produce your own example *.dvi or *.pdf file; rename the template file as you see fit and use it for your own input.

2. Step-by-Step Instructions

The following sections give you detailed instructions on how to set up your files and meet EDP's specific style and layout requirements. Please try to adhere to these standards right from the start and use them as a checklist before submitting the manuscript or print data.

These are the special commands we should follow before `\maketitle`:

`\institute` — Address

`\abstract` — Abstract

`\keywords` — Keywords

`\subclass` — Classifications

`\abstracttrans` — French abstract

`\keywordstrans` — French keyword

- English text convert to French text option `frenchtitle` in `\documentclass[...frenchtitle,...]{edpsci}`.

2.1. Initializing the Class

To format a *edpsci* enter

`\documentclass{edpsci}`

at the beginning of your input. This will set the text area to a `\textwidth` of 180 mm and a `\textheight` of 234 mm plus a `\headsep` of 12 pt (space between the running head and text).

N.B. Trim size (physical paper size) is 210 × 277 mm.

2.2. Required Packages

The following selection in the past has proved to be essential in preparing a fully formatted (i.e. ready-to-print) manuscript.

Invoke the required packages with the command

`\usepackage{}`

`graphicx.sty` is a powerful tool for including, rotating, scaling and sizing graphics files (preferably *.eps files)

2.3. Fine-Tuning Your Text

As a general rule, text, formulae, figures, and tables are typed using the standard L^AT_EX₂_ε commands. The standard sectioning commands are also used.

Always use the L^AT_EX commands `\label` and `\ref` for cross-referencing to sections, theorems, equations, figures, and tables. In contrast to any hard-coded references these soft-coded cross-references can automatically be converted to hyperlinks for any possible electronic version of your book.

Theorem-Like Environments. For individual text structures such as theorems, definitions, etc., the EDPSCI document class provides numerous predefined environments (*numbered* as well as *unnumbered*) which conform with the specific EDP Sciences layout requirements.

1. To produce a simple hyphen, used to connect or separate dependent parts of a word such as prefixes, or in compound adjectives, please enter a single keyboard hyphen without any space on either side (-).
2. To produce an en-dash, enter two single hyphens with no space on either side to stand in place of “to” in ranges, as in “Fig. 3a–c” or “... in the range 10–20 eV”, or to connect two names or words that are independent of each other, such as “... the electron–photon interaction”. However, double-barrelled names like Levi-Civita are connected with simple hyphens.
3. To produce an em-dash—e.g. to denote an insertion within a sentence—please enter three hyphens without any spaces on either side (---).

Quotation Marks. Please use the following commands to create English-language quotation marks: `'word'` gives ‘word’ in the dvi file, and `"word"` gives “word” in the dvi file.

Page Breaks. Please see to it that you always have at least two lines of the same paragraph at the foot or head of a page. So-called “orphans” or “widows” reduce the readability of your text.

Cross-References Within Text. Please always give a `\label` where possible and use `\ref` for cross-referencing. Such cross-references may then easily be converted to hyperlinks in any electronic version of your book.

The `\cite` and `\bibitem` mechanism for bibliographic references is also obligatory.

Cross-references to particular sections, figures, tables, equations and the like should be written in full when they stand at the beginning of a sentence, but in any other position within the text they should be abbreviated as follows:

(Section) Sect./Sects. (Figure) Fig./Figs.
(Page) p./pp. (Volume) Vol./Vols.

2.4. Fine-Tuning Mathematics

As a general rule, text, formulae, figures, and tables are typed using the standard $\text{\LaTeX}2_\epsilon$ commands. The standard sectioning commands are also used.

Always give a `\label` where possible and use `\ref` for cross-referencing. Such cross-references may then be converted to hyperlinks in any electronic version of your book.

Please set *mathematical expressions and formulae within the running text* in math mode, i.e. $\$ \dots \$$, so that the desired spaces are set automatically. In text mode please put a small space `\,` between a number and its unit.

Displayed Formulae will automatically be centered.

Equation Arrays. In order to get a readable layout for your equation arrays we recommend that you use the $\text{\LaTeX}$ environment `eqnarray`. This will automatically use optimal line spaces and line breaks. If an equation spans more than one line place the equals sign at the beginning of the second (or subsequent) line(s); binary operators such as $+$, $-$, $*$, etc. should also appear at the beginning of the second or subsequent lines of an array, and the line should be indented to the right of the equals sign in the line before.

Please *punctuate* displayed equations in the same way as any other written statement and insert `\;` before the punctuation to add a little extra space.

Multiplication. Where a multiplication sign is essential use the command `\times` ($\times$), not `\cdot` ($\cdot$). The `\cdot` is reserved for vector dot products.

Computer Code. To display computer code in your article, we recommend the use of the `verbatim` environment.

Variables should be represented by a unique single character and always, i.e. in math mode as well as in the text, be set in italics. If possible please use `\varepsilon` for ϵ and `\varrho` for ρ .

Subscripts and superscripts should always appear upright (use `\mathrm{ }` in math mode) when they are abbreviations. If you need to depict a vector, please also use the syntax `\vec{ }`. The font size will automatically be adjusted.

Fractions in displayed equations should be coded with `\frac`. When they appear within exponents, running text or narrow tables, they should be set with a slash. Otherwise the font size will be too small to be easily read.

Delimiters should be large enough to completely enclose their content – but no larger. We recommend using dynamic $\text{\LaTeX}$ input commands, e.g. `\left[` or `\right]`, `\langle` or `\rangle`, `\left|`, `\right|`, etc.

2.5. Figures, Tables and Their Captions

In general, text, formulae, figures and tables are typed using the standard $\text{\LaTeX}2_\epsilon$ commands. The standard sectioning commands are also used.

Always give a `\label` where possible and use `\ref` for cross-referencing. Such cross-references may then be converted to hyperlinks in any possible electronic version of your article.

Grids and details within the figures must be clearly readable and may not overlap.

Figures should be in *eps format* with fonts embedded, without preview and with the so-called bounding box adjusted to the actual content of the figure. Use the standard $\text{\LaTeX}$ “`graphicx`” package to include your graphics files.

Tables. By default, tables and their captions are justified. Please make sure that every table is *preceded* by a caption.

2.6. Special Elements

In the EDPSCI document class we have defined a few environments. This is done using the following commands

<code>\supplementary</code>	Supplementary Material
<code>\appendix</code>	Appendix
<code>\glossary</code>	Glossary
<code>\acknowledgements</code>	Acknowledgements
<code>\authorcontrib</code>	Authors Contributions
<code>\funding</code>	Funding
<code>\dataavailability</code>	Data Availability Statement
<code>\conflict</code>	Conflict of Interest
<code>\ethics</code>	Ethical Approval
<code>\informedconsent</code>	Informed Consent

2.7. Important Notes for the Production Process

Authors can support and speed up the production process by following the below described instructions.

General instructions

- Please make sure that any package used in the LaTeX file is compatible with MikTeX2.9/TeXlive2021 or higher.

Package

Algorithm

To typeset algorithms or pseudocode in LaTeX you can use one of the following options:

- Choose ONE of the (`algpseudocode` OR `algcompatible` OR `algorithmic`) packages to typeset algorithm bodies, and the algorithm package for captioning the algorithm.
- The `algorithm2e` package.

Important: Please choose only one of the above groups of packages, and use only the commands and syntax provided by the package you choose. These packages cannot be loaded simultaneously; otherwise you will get lots of errors.

2.8. References

References may be *cited* in the text either by number (preferred) or by author/year.

In mathematical texts references are often labelled as author-year acronyms. In order to achieve this simply give an optional argument to the `\bibitem` command. Always use `\bibitem` and `\cite` for cross-referencing.

When producing your bibliography please make sure that the data is complete (name and initial of author, year of publication, book title, publisher's name and place, journal name, volume number, page numbers) and up to date (e.g. edition number).

Please make sure that, in the individual reference citations, the initials of names do not stand alone. Please connect them to their surname with the help of the tilde `~` so that they will not be separated from each other when L^AT_EX breaks the line. The same applies to volume or page numbers.

3. Submitting your Manuscript

Please direct any queries concerning your book project to your contact person at EDP Sciences. He or she will be happy to respond directly or pass on your query to the expert in charge.